BANKING LOAN RISK ANALYSIS

Dataset Documentation

Capstone Project | Data Source, Dictionary, Quality, Transformation and Storage

Dataset: Default of Credit Card Clients (UCI Machine Learning Repository)

Scope: historical credit-card repayment records used for descriptive analytics and deterministic rule-based risk assessment; no AI/ML model is used.

1. Dataset Overview

The project uses the UCI Default of Credit Card Clients dataset, which records credit limits, demographic attributes, repayment statuses, monthly statement amounts, payment amounts and a historical next-month default outcome. The source contains 30,000 customer observations. The original analytical variables are X1–X23 with outcome Y and an identifier (ID), totaling 25 fields after header normalization.

Property	Documented value
Observations	30,000 customers
Original schema	ID, X1–X23, Y (25 columns)
Target/outcome	default payment next month: 1 = default, 0 = no default
Observation grain	One customer per source record
Time window	Six monthly repayment, bill and payment observations (September through April in project mapping)
Usage	ETL, portfolio analysis, and explainable rules-based risk segmentation

Source attribution: UCI Machine Learning Repository, Default of Credit Card Clients dataset. The dataset is commonly attributed to Yeh and Lien (2009). Original monetary units follow the source dataset; the application does not imply they are Indian rupees.

2. Original Data Dictionary

Field(s)	Meaning
ID	Unique customer identifier
LIMIT_BAL (X1)	Credit limit
SEX (X2)	Gender category code
EDUCATION (X3)	Education category code
MARRIAGE (X4)	Marital-status category code
AGE (X5)	Age in years
PAY_0, PAY_2–PAY_6 (X6–X11)	Monthly repayment status (six observations)
BILL_AMT1–BILL_AMT6 (X12–X17)	Monthly billed statement amounts
PAY_AMT1–PAY_AMT6 (X18–X23)	Monthly payment amounts
Y	Default payment next month: binary outcome

Repayment status uses source codes, including negative and zero values. Codes should be interpreted using the source documentation rather than treating all nonzero values as delinquency.

3. Extraction and Standardization

The input is an Excel (.xls) file. Extraction normalizes the source header row and maps the original X1–X23/Y variables to descriptive names, including ID, LIMIT_BAL, SEX, EDUCATION, MARRIAGE, AGE, PAY_0, PAY_2–PAY_6, BILL_AMT1–6, PAY_AMT1–6, and default payment next month. Raw values are retained for validation before curated

features are computed.

4. Data Quality Checks

The recorded validation run passed missing-value checks, duplicate-row checks, missing/duplicate customer IDs, credit-limit checks, gender checks, age checks and repayment-status checks. Two domain-value checks produced warnings rather than record rejection.

Check	Recorded result / treatment
Completeness and duplicate integrity	PASS: no missing values, duplicate rows or duplicate/missing IDs reported
Credit limit, gender, age, repayment status	PASS: no invalid values reported by the project validators
Education category	WARN: 345 records use undocumented category codes; mapped to Others for reporting while preserving original code
Marital-status category	WARN: 54 records use undocumented category codes; mapped to Others for reporting while preserving original code

These are recorded outcomes from the project validation run, not a guarantee that future incoming batches will pass. Preserving original category codes supports traceability and reprocessing.

5. Curated Features and Transformations

The curated dataset expands the extracted 25 fields into approximately 36 columns. It adds standardized names, descriptive demographic labels, six-month behavioral aggregates and rule-input features.

Feature / group	Definition or purpose
customer_id, credit_limit, age	Normalized identifiers and numeric attributes
gender, education, marital status	Original codes plus human-readable reporting labels
repayment_status_[month]	Six monthly repayment-status observations
bill_amount_[month]	Six monthly statement balances
payment_amount_[month]	Six monthly payments
delay_month_count	Number of months flagged with a payment delay
max_delay_status	Maximum recorded delay status
has_payment_delay	Boolean indicator for any detected delay
total_bill_amount_6m / average_bill_amount_6m	Six-month bill aggregates
total_payment_amount_6m / average_payment_amount_6m	Six-month payment aggregates
latest_utilization_ratio	Latest billed amount relative to credit limit
default_next_month	Historical observed outcome, retained for descriptive comparisons

The repayment month suffixes follow the project's September-to-April mapping. The dataset provides six observations, not a complete continuous calendar time series.

6. Rule-Based Risk Assessment

Risk assessments are derived from repayment frequency, repayment severity, utilization and payment-to-bill behavior. The scoring system is deterministic, interpretable and does not train a machine-learning model. The historical default label is not an input to the score.

Risk component	Possible points
Delay frequency	0–3
Delay severity	0–3
Credit utilization	0–3
Payment behavior	0–2
Total risk score	0–11

Risk band	Score range
LOW	0–2
MEDIUM	3–5
HIGH	6–11

Recorded baseline distribution: LOW 13,966 customers; MEDIUM 9,673; HIGH 6,361 (total 30,000). These counts describe the processed historical baseline, not a forecast of future defaults.

7. MySQL Storage and Reconciliation

The relational warehouse separates customer demographics, credit profiles and month-level activity. A separate table holds rule-based risk assessments, while ETL batch audit records document incremental processing.

Table	Role / baseline row count
customers	Customer identities and demographic attributes — 30,000
credit_profiles	Credit and derived behavioral profile — 30,000
monthly_credit_activity	Six monthly observations per customer — 180,000
customer_risk_assessment	Deterministic risk components and bands — 30,000
etl_batch_audit	Per-batch source/new/updated/unchanged/failed counts and status
dim_gender, dim_education, dim_marital_status	Category lookup/dimension tables

Initial load reconciliation passed for the 30,000 / 30,000 / 180,000 customer/profile/activity counts. Incremental processing distinguishes new, changed and unchanged customers; audit results record committed or rolled-back batches. One no-change batch processed 30,000 unchanged rows.

8. Limitations and Responsible Interpretation

This is a historical credit-card dataset, not live banking transaction data. It should not be represented as a loan-approval system or as a validated probability-of-default model. Risk bands are transparent rule-based segments intended for analytical exploration, not automated credit decisions. The outcome label represents historical next-month default and must not be described as a model prediction.

Demographic variables are retained for descriptive segmentation, but sensitive attributes require caution and appropriate governance. The dataset is not evidence of fairness, calibration, causal effects or generalization to other countries, lenders or time periods.

9. Reproducibility and References

Relevant project modules: etl/extract.py, etl/validate.py, etl/transform.py, etl/load.py, etl/incremental_load.py, rules/risk_rules.py, rules/load_risk.py, and database/schema.sql. For exact transformations and thresholds, refer to the version-controlled implementation and docs/risk_methodology.md.

External source: UCI Machine Learning Repository — Default of Credit Card Clients. Repository page: https://archive.ics.uci.edu/dataset/350/default+of+credit+card+clients

This document reflects the recorded project baseline and known implementation details. Re-run validation and row-count reconciliation if the source file or transformation code changes.